

SUMMER INTERNSHIP PROJECT REPORT

PlacePro AI: Predict. Prepare. Get Placed.

An AI & IoT Powered Multidimensional Platform for Student Placement Readiness, Intelligent Resume Screening, and Behavioral Analytics.

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| Executive Summary

Internship Context

The six-week summer internship (June 1 - July 10, 2026) was conducted at the IoT and Drone Design Lab at MMMUT, focusing on integrating advanced software technologies (AI/ML/NLP) with physiological hardware sensors.

The collective research led to the conception and development of **PlacePro AI**, bridging the gap between passive learning and career launch capabilities.

PlacePro AI Concept

An intelligent ecosystem built to evaluate employability. Rather than basic keywords, it evaluates technical skill fit, code profile strength, communication clarity, and physical interview posture/stress.

Leverages robust ML regressors for reliable ATS score generation, alongside Generative LLMs for customizable 30-to-90-day learning roadmaps.

| Background & Industry Need



AI-Driven Recruitment

Modern HR systems automate up to 90% of screening. Initial filters reject profiles before human review. Machine Learning is now mandatory to benchmark candidate capabilities accurately.



ATS Software Need

Applicant Tracking Systems (ATS) reject solid profiles due to parsing flaws or keyword mismatches. Students need predictive diagnostic platforms to test their format compatibility pre-submission.



The Missing Link

Existing platforms evaluate resumes in isolation. They fail to test live behavioral communication, stress indicators, or construct dynamic learning roadmaps to bridge skill gaps.

| Problem Statement



Sub-optimal Resume Formatting

Qualified candidates get filtered out at early stages because of ATS-incompatible visual templates or layout parsing errors.



Invisible Skill & Keyword Gaps

Students fail to realize that key tech stacks or certifications requested by specific target companies are completely absent from their profiles.

Inability to Gauge Readiness

A complete lack of data-driven, unified preparation benchmarks combining technical skill scores with physical interview behavioral feedback.



Stressed student looking for placement guidance

| Core Project Objectives



Automatic Parsing & Extraction

Parse complex structures of PDF resumes to accurately segment technical skills, projects, and experiences into clean structured models.



ML-Based ATS Diagnostics

Train robust regression models to accurately predict the candidate's ATS probability and outline actionable improvements.



Integration of Code Profiles

Consolidate platform activity such as LeetCode performance and GitHub commits into a unified readiness scorecard.



IoT-Assisted Stress Tracking

Design an ESP32 hardware node with biometric pulse and spatial sensors to monitor emotional and physical indicators during interview simulations.



Dynamic Roadmap Generation

Provide automated, custom-tailored weekly learning roadmaps powered by Large Language Models to systematically build target skills.

| Literature Survey & Research Gap

📖 Existing Tech & Studies

Prior systems rely strictly on basic keyword frequency counters or simple boolean checks, neglecting semantic sentence context.

Coding platforms (e.g. HackerRank) or mock setups operate in silos—forcing students to juggle multiple disparate tools with no single point of assessment.

⚠️ Identified Research Gap

No existing commercial or academic system captures the physical or emotional components (posture, movement, pulse fluctuations) during mock oral tests.

Feedback reports remain entirely static and rigid. Dynamic, personalized career roadmaps that evolve based on test results are completely missing.

| System Architecture

⚙️ Modular Technical Flow

Frontend layer: React.js designed with Tailwind CSS to offer a sleek, modern dashboard showcasing key performance metrics.

Backend API: Built via FastAPI (Python) implementing headless async requests. Secure user database queries are handled in PostgreSQL/SQLite.

AI Analytics: Integrates Google Gemini API for resume diagnostic intelligence alongside OpenCV and MediaPipe for facial & physical analysis.



| Core Platform Modules



1. Resume Intelligence

Automated parsing using NLP libraries. Predicts accurate ATS scores and generates refined professional summaries.



2. Code & Aptitude

Integrates with API endpoints to pull public metrics. Incorporates structured quantitative and logical tests.



3. Mock Interview

Performs voice-to-text processing using Whisper API. Evaluates responses based on structured company criteria.

4. Behavioral IoT

Utilizes ESP32 nodes to track live pulse metrics, posture, and nervous restlessness throughout mock sessions.

IoT Hardware & Sensors

Physical Node Assembly

To record physical stressors during interview preparation, an auxiliary biometric and spatial tracking device is integrated into the workspace.

MAX30102 Heart Rate Sensor

Measures physiological pulse rates and detects nervous cardiovascular spikes in real time.

MPU6050 Accelerometer & Gyro

Detects nervous restlessness, micro-movements, or severe forward slumping on the chair.

ESP32 Microcontroller

Streams physical metrics over Wi-Fi sockets directly to the FastAPI server.

 ESP32	 Programming Cable	 Breadboard	 OLED	 Motion Sensor
 DHT11 Module	 LDR Module	 IR Sensor	 Relay Module	 Buzzer Module
 10k Pot	 Push Button	 Red Led	 Green Led	 Yellow Led
 RGB Led	 220 Ohm	 330 Ohm	 1k Ohm	 10k Ohm
 F-M Jumper	 M-M Jumper	 F-F Jumper	 IoT CLOUD	 Resistor Color Card
				

| System Data Flow Timeline



Machine Learning Evaluation

ML Algorithm Tried	MAE (Mean Abs Error)	R ² Accuracy Score	Project Decision
Gradient Boosting Regressor	2.27	0.9831	✔ Selected for Deployment
Random Forest Regressor	2.45	0.9724	Lower performance on testing data

Testing Dataset: AI Resume Screening dataset containing 1,000 resume records with 11 core profiles attributes (skills, experience, certifications).

📊 Error Metrics Formula

The models were optimized based on Mean Absolute Error (MAE) minimizing prediction error variance:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

| Feasibility & Social Impact

Project Feasibility

Technical Feasibility: Employs robust standard APIs (Gemini, FastAPI) requiring low compute overhead. High operational modularity.

Economic Feasibility: Utilizes accessible micro-sensor hardware costing under 3,000 INR. All core software dependencies are open-source.

Societal Benefit

Democratizing Placement Prep: Grants premium level technical critique to students lacking access to expensive training camps.

Confidence and Readiness: Promotes self-paced correction of critical skill gaps, drastically dropping post-graduation unemployment spikes.

| Conclusion & Future Scope



Project Conclusion

- **Integrated Environment**
Successfully built and demonstrated a full-stack platform aligning machine learning analytics with behavioral parameters.
- **High Metric Accuracy**
Achieved reliable predictive results utilizing the Gradient Boosting Regressor model (R^2 Score: 0.9831).

Future Enhancements

- **AI Interview Simulators**
Integrating conversational avatars mimicking recruiter behavior dynamically in real-time response setups.
- **Advanced Wearables**
Moving beyond desktop sensor boards to seamless wearable wristbands monitoring bio-signals across longer simulation runs.

Questions & Answers

Thank you for your valuable time and attention!

IoT and Drone Design Lab — Department of ECE

Madan Mohan Malaviya University of Technology, Gorakhpur

Academic Year Session: 2026-2027

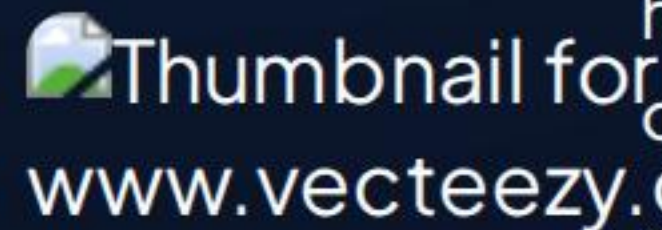
"We hereby declare that this presentation represents our original work carried out during the six-week summer internship program on IoT, Drone, AI and ANSYS modeling."

| Image Sources

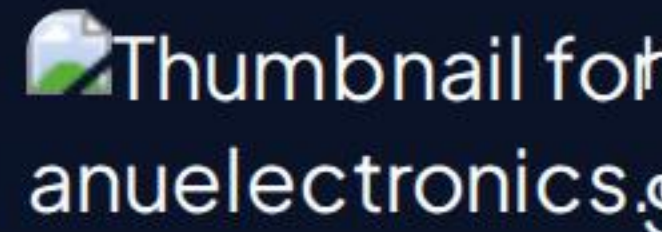


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